HexManiacAdvance

Mac Edition

Updated November 9, 2026

Development accomplishments

81 groups / 24,292 assertions / 0 failures / 0 skipped groups

Last complete Mac regression: November 8; subsequent font controls passed separate focused checks.

UI Improvements

- **Mac Edition welcome screen** - Added a responsive welcome screen with a faded Tutorial Village background, contributor links and a prominent Open ROM button.
- **Readable script display** - Aligned commands, dialogue, comments and highlighting with a shared monospaced font and corrected placeholder spacing.
- **Compact map tutorials** - Made tutorial cards readable on hover, reduced their footprint and kept map previews behind editor overlays.
- **Continuous encounter editing** - Kept focus and caret position stable while editing grass, surf, tree and fishing encounter tables.
- **Quick field replacement** - Made Tab and Shift Tab select complete short-field values without selecting entire script or dialogue editors.
- **Stable sprite selection** - Fixed large NPC sprite previews and recovered selector state after scrolling, history changes and reopening.
- **Fresh portrait and palette views** - Refreshed edited portrait choices and dismissed stale palette editors after Undo, reload or context changes.
- **Simpler toolbar** - Hid optional Claude and built-in emulator controls while retaining their implementations and external emulator access.
- **Readable selected source** - Made script selections translucent so syntax-coloured text remains readable while highlighted.
- **Unobstructed editor panels** - Reserved scrollbar space so hovering cannot cover event arrows, tiles or other editor controls.
- **Explicit script text sizing** - Replaced accidental Command-scroll resizing with minus/plus controls and a size indicator.

Mac Port Fixes

- **Control-click context menus** - Restored existing menus for tabs, warps, map arrows, hex content and palettes using Mac secondary-click conventions.
- **Finder integration** - Replaced Windows file-location launching with Finder reveal for the owning ROM file.
- **Remembered emulator choice** - Made Play launch a selected external emulator such as mGBA, remembering its application path and supporting Browse.
- **Focused clipboard commands** - Kept Command copy, cut and paste on the focused text or palette target instead of falling through to binary editing.
- **Safe field shortcuts** - Kept Delete, Backspace and arrow keys inside focused map-event fields instead of deleting or panning the event.
- **Control-click eyedroppers** - Made Control-click sample sprite colors and block tiles without entering the painting path.
- **Palette color entry** - Restored propagation of named swatch colors and hexadecimal color edits into palette data.
- **Map entry stability** - Removed the cold-map UI-thread error and focus-outline animation crash.
- **Restored editor controls** - Fixed rotated map arrows and restored specialized trainer, sprite and palette templates.
- **Ready-to-type Go To** - Focused the search field after Command-G, including after opening a ROM or changing tabs.

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Map Editing Improvements

- **Trackpad map panning** - Added Option-drag panning that preserves the current brush and finishes correctly when Option is released first.
- **Placed-block selection** - Restored Mac Control-drag selection of placed blocks while preserving their collision and elevation data through saves and history.
- **Brush placement preview** - Added a translucent preview of the selected block pattern at its stamp destination.
- **Mixing bag instructions** - Corrected Mac help for Command-click bag membership, clearing the bag and activating random painting.
- **Permission-safe painting** - Preserved movement permissions during bag flood-fill and single-block rectangle painting.
- **New object scripts** - Restored the script-creation button so an object can receive an allocated starter script and attached pointer.
- **Stable NPC previews** - Eliminated complex hover-preview flicker and refreshed summaries after dialogue edits, Undo and Redo.
- **Interrupted gesture cleanup** - Stopped ghost painting and panning after capture loss, and kept interrupted strokes separate in Undo history.
- **Reliable same-cell edits** - Allowed painting, brush replacement and permission changes at the same cell after Undo or Redo.
- **Reveal mode cleanup** - Cleared transient Space reveal when keyboard focus, tabs or editing contexts change.
- **Trackpad-friendly zoom buttons** - Added centered minus and plus buttons that use the existing zoom levels and disable at their limits.
- **Map dimensions at a glance** - Displayed map size in tiles, native pixel dimensions on hover and clearly labeled cursor coordinates in the status bar.
- **Direct map PNG export** - Saved map images to MapImages beside the ROM with Overwrite, Save Updated Copy and Cancel choices for existing files.
- **NPC template sprite choice** - Added a bounded pre-placement sprite picker that remembers its independent choice and toolbar icon for the open ROM session.
- **Bounded tileset previews** - Kept blockset previews within the map viewport and limited standalone rendering to valid block counts.
- **Stable tileset selection** - Fixed the map-header dropdown crash while preserving immediate pointer writes, fresh previews and Undo/Redo.
- **Safe border access** - Rejected invalid border coordinates before calculating ROM addresses to protect neighboring data.
- **Consistent nine-grid permissions** - Applied explicit movement-permission choices to both inferred and fallback brush corners.
- **Stationary block information** - Gave Block ID a fixed position beside Edit Block without overlapping movement controls.
- **Reachable tile flip controls** - Kept flip arrows visible and clickable while crossing the gap beside each block quadrant.
- **Accurate header script rows** - Refreshed conditional map-script rows after adding or deleting entries while preserving Undo/Redo.
- **Cleaner export confirmation** - Displayed the saved filename and folder in a compact confirmation with Reveal in Finder and OK actions.

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Scripting Improvements

- **Automatic dialogue scaffolds** - Inserted an automatic text pointer and complete dialogue block with the caret ready for typing.
- **Automatic movement scaffolds** - Added complete movement blocks for player, camera, NPC and applymovement commands when the required operands are available.
- **Completion-aware scaffolding** - Ran dialogue and movement assistance after accepting a command suggestion as well as typing the command directly.
- **Flag and variable assistance** - Restored existing unused-ID helper buttons and Command U routing for eligible flag and variable commands.
- **Accurate autocomplete placement** - Corrected caret mapping and command replacement across multiline scripts while preserving explicit hexadecimal values.
- **Visible dialogue warnings** - Made overflow warnings and script errors visibly underline the affected text without adding popups or automatic wrapping.
- **Usable suggestion lists** - Fixed mouse acceptance and dismissed stale suggestions after outside clicks, focus changes, detachment or context replacement.
- **Aligned trainer warnings** - Corrected trainer move-token validation and warning spans while preserving existing matching and legality rules.
- **Map reference beside source** - Added a separate Map View with panning, zoom and an optional light tile grid while keeping tiles uneditable.
- **Click-to-open event scripts** - Opened object, script-event and script-signpost code from Map View while protecting incomplete drafts.
- **Useful event context** - Displayed Object ID and assigned Flag or none for objects, plus Variable Number for script events.
- **Steady long-script editing** - Stopped viewport jumps caused by focus changes and delayed compilation refreshing another script section.
- **Replace sign dialogue directly** - Selected only the first dialogue block when entering through a signpost arrow, with ordinary clicks clearing selection.
- **Reliable initial assistance** - Handled typed and accepted commands, trailing whitespace and earlier pointer edits without overwriting explicit addresses.
- **Fresh unused IDs** - Updated flag and variable availability after edits, deletion and history changes while retaining protected reservations.
- **Earlier overflow indicators** - Added a six-pixel advisory cushion to dialogue warnings without changing text encoding or inserting breaks.
- **Clean completion dismissal** - Kept accepted movement suggestions closed and prevented truncation markers from entering source.

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Under the Hood Improvements

- **Safer file saves** - Staged file replacements and preserved open unsaved documents when ROM or metadata saving fails.
- **Validated binary patches** - Validated patch structure and UPS checksums before changing ROM bytes, preserving existing edits after rejected IPS or UPS imports.
- **Validated project imports** - Rejected incompatible layouts, invalid values and lossy text before applying edits, with rollback if an import fails.
- **Recoverable project exports** - Published consistent table exports with owned-file manifests, writer coordination and recovery from interrupted publication.
- **Safer external reloads** - Handled replacement and rapid external saves more reliably while protecting local unsaved work.
- **Protected pending script source** - Bound deferred edits to their original context and accounted for pending source during Save, Save As, Close and reload.
- **Atomic script compilation** - Rolled back failed script and resource compilation across bytes, pointers, relocation, metadata and history while retaining drafts and diagnostics.
- **Mac diagnostics and updates** - Pointed crash/help links and compatible-release checks at the renamed Mac Edition repository while retaining upstream references where appropriate.
- **Gap-safe script decompilation** - Kept referenced scripts separate across unknown byte gaps instead of emitting uncompilable raw FF source.
- **Signpost relocation safety** - Updated script-bearing signpost targets when entry code moves, including failure rollback and exact Undo/Redo.
- **Resolved draft warnings** - Cleared an owned rejected-draft warning after correction without clearing newer or unrelated errors.
- **Reliable preference reloads** - Preserved unusual recent-file names and restored search limits independently of diff limits.
- **Safer first launch** - Handled settings-access failures and rejected invalid ROM metadata without aborting startup.
- **Repeatable Mac packaging** - Built architecture-specific app bundles without deleting the last successful output when packaging fails.
- **Tutorial patch framework** - Prepared a disabled-until-ready action that validates the English FireRed 1.1 base and creates a separate patched ROM.

Verification Improvements

- **Complete Mac regression runner** - Passed 81 groups and 24,292 assertions with zero failures and zero skipped groups in the November 8 complete Mac run.
- **Real editor workflow coverage** - Expanded synthetic UI coverage for gestures, clipboard routing, suggestions, image previews, persistence and history.
- **Portable engine test hosts** - Enabled the original unit and integration test hosts on Mac with explicit optional-ROM skips and recorded legacy expectation differences.
- **Latest focused checks** - Passed seven related groups and 1,024 assertions for the subsequent font-control change without claiming another complete-suite run.
- **Reproducible release preparation** - Recorded source-install and helper-package rehearsals plus a checklist separating completed checks from distribution gates.
- **Recoverable development history** - Kept fixes and investigation evidence in separate commits with documented checkpoint tags for later review or rollback.

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Optimizations

- **Script highlighting** - Reduced the recorded 2,000-line highlighting workload from about 197 to 7.8 ms and temporary allocation from 21.6 to 3.3 MB per edit.
- **Script argument lookup** - Reduced a 2,000-command deferred update from 2,795 to 90 ms in the repeated-argument benchmark.
- **Script width layout** - Coalesced ten queued width calculations into one, reducing the measured 500-line burst from 39.6 to 3.3 ms.
- **NPC hover lookup** - Reduced empty-tile lookup on the 32-NPC fixture from about 4.51 to 0.065 ms by avoiding unrelated event-editor construction.
- **Map zoom rendering** - Removed all 70 unnecessary bitmap publications in the zoom-only benchmark while preserving rendered pixels.
- **Table navigation** - Reduced the measured binding and layout time for 100 complex navigation switches from 924 to 675 ms.
- **Portrait preparation** - Reduced 100 candidate-preparation operations from 77.4 to about 3.0 ms while preserving image freshness.
- **Trainer move validation** - Reduced validation of 100 six-member custom teams from 83.1 to 15.0 ms.
- **Item usage searches** - Reduced a measured delayed map-usage query from 225 to 148-160 ms by reusing script outcomes within that query.
- **Released detached editor controls** - Reduced settled managed heap from 225.7 to 61.2 MiB in Hex mode and 212.1 to 62.3 MiB in the paired Map tab-switch workload.
- **Unused-ID graph scanning** - Reduced initialization from about 1,342 to 4.6 ms and post-edit refresh from 607 to 2.2 ms in the synthetic shared-tail workload.

How to read the metrics

Figures describe recorded component or synthetic workloads, not an overall app speedup; timings cannot be added, and temporary allocation differs from retained managed heap and native RAM.

Evidence: CodexChanges entries 001-213, the performance reports under tests, and tests/MAP_REFERENCE_MERGE_VERIFICATION.json for the complete run at b6b642f9; later font-control checks are recorded separately in entry 213.

The Tutorial Village background is included; the playable tutorial patch and guides remain in development, and HMA patch compatibility with the correct base ROM still awaits user verification.

Built on Haven's HexManiacAdvance and the original Mac port by xavierkeys; Mac Edition is maintained by arkitekt13.